The KERAAL project in plain words

What we are doing, why, and where we are now

In one sentence. We study whether a computer can watch a rehabilitation exercise through a camera and tell if it was done correctly. What we found so far: **the computer can partly recognise wrong movements. The hard part is that the line between "slightly off" and "an error" is drawn differently by different data sets and different experts.**

1. Why this study

- People with low back pain must do rehabilitation exercises at home, alone. Nobody tells them when they do it wrong.
- A camera app that says "raise your arms higher" would help. Our goal is to check with data whether such an app can work.

2. The data: KERAAL

A public data set from a French research team. A Kinect camera recorded the positions of 25 body joints (the "skeleton") 15 times per second.

Exercise	What the person does	Typical error
CTK	Raise both arms to hide the face	Arms not raised enough
ELK	One arm over the head, bend the body sideways	Body not bent enough
RTK	Turn the upper body to the side	Not turned enough

Group	Who	Answer (label)
1A	6 patients	Two physiotherapists each judged the videos
2A	6 healthy people	Same as above
3	3 healthy people acting out errors on purpose	Written in the file name (C = correct, E1-E3 = error)
1B, 2B	Extra recordings	No labels

3. The main question

Labelling patient data by experts is slow and expensive. Data where healthy people **act out errors (Group 3)** is cheap and plentiful. **Can a computer that learned from acted errors recognise real patients' errors?**

4. What we did and found

Step	What we did	Result
1. Build	Built the programs that read, clean and evaluate the data (the pipeline)	Done and checked
2. Pre-registered test	Fixed the rules in advance, then trained on acted data and tested on patients	**Did not work**: the computer called almost every patient movement (92-97%) an error
3. Why?	Looked closely at the data	Acted errors are exaggerated. The two experts use different lines (about 12 degrees apart on torso rotation). Errors cluster in a few people
4. Opposite direction (teammate report)	Trained on real data, tested on acted data	**Worked**: wrong movements got the higher score 71-77% of the time (chance = 50%)

What it means. The computer can roughly rank which movement is more wrong. It failed because **where " wrong" begins (the line)** differs between acted data, patients and experts. This is likely the central story of our paper.

5. Who does what now

Person	Role	Current work
Team lead	Direction, writing	Coordinates the team, shapes the paper's story
Uijin	Data	Cleaned data v2, error-segment table, extra data
Roshini	Programming	Tests whether re-setting only the line restores performance
Joshi	Understanding	Understand the project with this sheet (below)

6. Words you will hear

Word	Meaning
Skeleton	Joint positions found by the camera: 25 points, not a video
Label	The answer: "this movement is correct / an error"
Annotator	A physiotherapist who labelled the videos (two: A and B)
Train / test	Data the computer learns from / new data used to check it
AUC	How often a wrong movement gets the higher score. 0.5 = coin flip, 1.0 = perfect
Chance level	The score you get with shuffled labels. A real result must beat it
Pre-registration	Fixing the hypotheses and method before seeing results, so we cannot cherry-pick
Exploratory	Extra analyses after the pre-registered one; for finding new ideas

7. Joshi's task

- Read this sheet (20 minutes). Ask about any word you do not understand.
- Explain the project to the team lead in **3-5 sentences**: what are we doing, and what did we find? (Speaking or a message is fine.)
- **Optional**: in the Joshi_videos folder, watch one correct video (C) and one error video and describe the difference in one line.